

OLOS: The Open Network for Customer Loyalty

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An open, neutral interoperability protocol and settlement network for consumer reward value.

OLOS establishes the missing protocol layer for customer loyalty. It acts strictly as neutral infrastructure — not a competing loyalty platform, consumer app, or proprietary exchange. By decoupling programmatic business rules from transactional settlement, OLOS allows tokenized loyalty value to flow safely across merchant boundaries while preserving 100% merchant data sovereignty and program control.

Why OLOS / Why Now?

1. Historical Context: Loyalty Needs Its Internet Moment

Modern digital commerce was unlocked by open, neutral protocols:

- **Email** scaled because **SMTP** provided a universal messaging standard that no single company owned.
- **The Web** exploded because **HTTP/HTML** established an open addressing and content retrieval system.
- **Payments** achieved global reach because networks like **Visa, Mastercard, SWIFT, and FedNow/RTP** abstracted away cross-bank value movement under standardized, secure clearing rails.

Loyalty was left behind.

Today, the average consumer manages 15 to 30 separate reward accounts. Every brand operates inside an isolated, proprietary silo. Because loyalty value cannot move cleanly across borders, the market suffers from massive economic deadweight loss:

- **Merchants** carry **\$100B–\$500B+ in unredeemed liabilities** on their balance sheets, facing low engagement (~22% average redemption rates) and high point-to-point integration costs.
- **Consumers** experience “fragmentation fatigue,” watching trapped rewards expire or go unused.

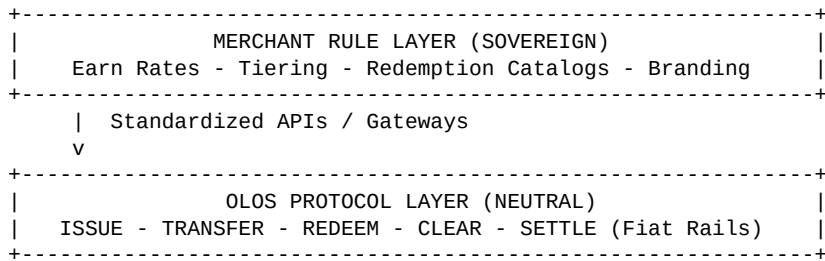
2. Why Legacy Approaches Failed

Prior attempts to fix loyalty fragmentation introduced fatal structural flaws:

Approach	Structural Flaw	Why It Fails
Coalition Schemes <i>(e.g., Plenti)</i>	Surrender of Sovereignty	Merchants lose direct customer relationships, branding, and first-party data to a central operator. ~60% fail within 10 years.
Point Exchanges	High Rents & Spreads	Extract 10%–30% transaction spreads with slow, batch-reconciled APIs rather than real-time settlement.
Web3 / Crypto Tokens	Volatility & UX Friction	Price volatility makes financial accounting impossible; gas fees, seed phrases, and legal ambiguities destroy consumer adoption.

3. The Core Innovation: “Separate the Rules from the Rails”

OLOS inverts previous designs with a single architectural principle: **decouple business logic from transactional infrastructure.**



- **The Rules Layer (100% Merchant-Owned):** Merchants retain full control over earn/burn ratios, promotional rules, customer communications, and customer data. Personally Identifiable Information (PII) **never** touches the shared ledger.
- **The Rails Layer (Neutral Infrastructure):** OLOS provides standardized transaction primitives (ISSUE, TRANSFER, REDEEM, CLEAR, SETTLE) and an automated multilateral netting engine that clears cross-brand obligations into sovereign fiat via instant banking rails (ACH, RTP, FedNow, SWIFT).

4. Why Now?

Three macro catalysts make this moment unique:

- 1 **Instant Payment Rails Are Mature:** Real-Time Payments (RTP), FedNow, and SEPA Instant allow automated, intraday multilateral netting without capital lockup or crypto volatility.
- 2 **Merchant Exhaustion:** Retailers and airlines are weary of closed, high-fee payment and marketing channels. They demand open standards that protect their data while increasing balance sheet velocity.
- 3 **Consumer Expectations:** Modern consumers expect the same liquid, friction-free experience from loyalty rewards that they experience with instant messaging and modern fintech apps.

The OLOS Thesis: Where Open Protocols Differ

Existing loyalty systems are closed ecosystems; legacy coalitions achieve interoperability by forcing merchants into centralized commercial and data-sharing relationships.

OLOS explores a different approach: **Can an open, neutral protocol deliver global interoperability while allowing each merchant to retain absolute sovereignty over its currency, rules, identity, and customer relationships?**

The OLOS architecture enforces this thesis through a strict 7-layer separation of concerns:

1. Merchant Sovereignty	- Local rules, earn/burn rates, PII isolation	
2. Trust Registry	- Identity verification & cryptographic keys	
3. Protocol Layer	- Primitives: ISSUE, TRANSFER, REDEEM, CLEAR	
4. Rules Engine	- Deterministic exchange & policy evaluation	
5. Clearing & Netting	- Automated multilateral obligation calc.	
6. Settlement	- Sovereign fiat resolution via RTP/FedNow	
7. Governance	- Open foundation, OIP process, dispute rules	

- **Merchant Sovereignty:** Merchants maintain complete control over their program economics, branding, and customer relationships. Zero PII leaves local systems.
- **Trust Registry:** Public-key infrastructure (PKI) and directory services establish trusted network participants without centralizing user data.
- **Protocol Layer:** Standardizes immutable transaction events and message semantics across disparate systems.
- **Rules Engine:** Evaluates exchange parameters and eligibility deterministically before value moves.
- **Clearing & Netting:** Aggregates multi-party transactions to minimize capital liquidity requirements.
- **Settlement:** Discharges netted obligations in fiat via regulated banking rails.
- **Governance:** Standardized, non-profit stewardship defining network entry, specifications, and dispute resolution.

Core Protocol Primitives

All value movement across the OLOS network is governed by five atomic, cryptographically signed primitives:

- **ISSUE:** A merchant Gateway creates tokenized loyalty value and credits it to a cryptographic identity.
- **TRANSFER:** Moves unencumbered tokenized value from one holder to another (peer-to-peer or consumer-to-merchant).
- **REDEEM:** A consumer presents value for fulfillment against goods/services; burns or locks claims on the ledger.
- **CLEAR:** Internal system primitive used during multilateral netting cycles to calculate net fiat obligations.
- **SETTLE:** Triggers and confirms fiat movement between merchant settlement accounts to discharge netting balances.

Repository Structure

```
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+-- docs/
|   +-- whitepaper/           # Core White Paper (Volumes I-VII)
|   +-- oips/                 # Open Improvement Proposals
|   +-- architecture/        # Sequence diagrams, topology specs
+-- spec/
|   +-- schemas/              # JSON / Protobuf definitions for ISSUE, TRANSFER, REDEEM
|   +-- state-machine.md      # Ledger state transitions & invariants
+-- gateway-reference/        # Open-source Merchant Gateway reference implementation
|   +-- src/
|   +-- docker-compose.yml
+-- test-vectors/             # Conformance test suites for Gateway integrators
```

Roadmap & Governance

OLOS is managed by the **OLOS Architecture Working Group**, a neutral, non-profit-aligned working group dedicated exclusively to standard maintenance, security audits, and open-source tooling.

- **Phase 1: Working Group & White Paper (Q3–Q4 2026)** — Working group establishment & founding merchant anchor cohort recruitment.
- **Phase 2: Protocol Specification (Q4 2026–Q2 2027)** — Release of technical specifications, SDKs, and Gateway reference builds.
- **Phase 3: Devnet & Sandbox (Q2–Q3 2027)** — Sandbox deployment for third-party developer testing.
- **Phase 4: Private Beta Consortia (Q3–Q4 2027)** — Pilot implementations with founding merchant cohort.
- **Phase 5: Public Testnet (Q4 2027–Q2 2028)** — Node operator bootstrapping and governance activation.
- **Phase 6: Mainnet Soft Launch (Q2–Q4 2028)** — Live production deployment on banking settlement rails.

License & Intellectual Property

- **Documentation & Specifications:** Licensed under Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0).
- **Code & Reference Implementations:** Released under permissive open-source licenses (Apache 2.0 / MIT).

Community & Contact

Website: olos.org (Upcoming)

GitHub: github.com/markangell88-olos/OLOS-Architecture-Working-Group